Chamber Literature Synthesis

What 69 held sources (50 papers plus standards, reports and grey literature) on anechoic chambers, rotor test facilities and free-field qualification say when read together: a graded evidence matrix, where they converge, where they genuinely disagree and why, what nobody has measured, and what the weight of evidence supports for the arc rig.

2026-09-07 · rev 2 (post-audit)
method: academic-research-skills / deep-research, lit-review mode
(bibliography → source verification → synthesis; matrix + convergence + tension inventory + gaps)
every cell traces to a quoted passage in papers/*/EXTRACTS-*.md or MATRIX-3W.md
source texts govern where this synthesis disagrees

Read one at a time, these papers are facility reports, standards arguments and rotor-noise studies with little in common. Read together they make one argument in nine parts. The wedge cut-off printed on a chamber's brochure is a property of the absorber, not of the room, and the room's real limit sits 20–200 % higher (T2). Whether you see that limit depends on how you look: pure tones, dense sampling and a fixed-slope fit reveal it, band noise and a free acoustic-centre fit hide it (T1, T3). The reflections that matter are rarely the walls; they are floors, doors, grates, stands and the measuring rig itself (T4, T7). Put a rotor in the room and a second, aerodynamic artefact appears within seconds (T5). Models built on absorption coefficients cannot see any of this, because the failure is a phase effect (T6). Extending the usable range is possible but every demonstrated route is small-scale or geometry-specific (T8), and open-jet tunnels trade room reflections for shear-layer scattering (T9).

01 Evidence matrix

Themes T1–T10 are defined in the convergence table below. Cells: **S** supports · **C** contradicts · **P** partial or mixed · — not addressed. Level: III controlled measurement with reference or qualification data · VI single case or simulation without measured validation · VII standard, vendor note or expert report. Quality A–D per the skill's evidence template (A peer-reviewed with full method and numbers … D abstract only). Full WHY/HOW/WHAT rows with quotes are in `papers/chamber-problems/MATRIX-3W.md`, `papers/anechoic-simulation/MATRIX-3W.md` and `papers/reflection-localization/MATRIX-3W.md`.

SOURCE	KEY FINDING (FROM THE EXTRACTS)	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	LVL	Q	SET
Aussal & Gueguen 2019, ICA Aachen	Complexity "quasi linear" (Fig. 3); validity condition ka ≫ 1 (Eq. 9); "as ray-tracing modelization is an high-frequency approximation …	—	—	—	—	—	P	—	—	—	—	VI	B	simulation
Bikmukhametov 2026 (arXiv 2603.16556, "submitted to Elsevier")	Disk α = 0.17–0.5; modelled pyramid α = 0.815 at 290 Hz rising to 0.996, >0.9 above 410 Hz. RT drops from 1.5 s at 63 Hz to 0.21 s at 100 Hz, then …	P	S	P	—	—	—	S	P	—	—	III	B	simulation
Chialina et al. 2015, arXiv 1511.08410	"In almost all cases the simulation lies within the uncertainty band." (§7.2, p. 16). "The distortion of the radiation pattern due to t…	—	—	—	S	—	P	P	—	—	—	III	C	simulation
Fonseca, Brum, Mareze & Brandão 2018, DAGA München	Windowing outperformed TDS ("ripples in FRF obtained via TDS (comb filtering effect)"). "It is important to notice that better results …	—	—	—	P	—	—	—	P	—	—	VI	B	simulation
González & Kob 2026, DAGA Dresden	"In the reverberant hall and laboratory at ETI, significant differences in sound pressure levels are observed, particularly for frequencies of 1 …	—	—	—	P	—	—	S	—	—	—	III	B	simulation
Götz et al. 2025, arXiv 2509.05175 (Treble Technologies)	ρ (measurement vs simulation): DG-FEM 0.73–0.92, GA-RR 0.57–0.77, GA-RT 0.10–0.56 (Table 3). "While numerical wave-based simulations yielded simi…	—	—	—	—	—	S	—	—	—	—	III	B	simulation
Huang (Univ. of Bristol, supervisor Azarpeyvand; year unclear in text,	"At H=0R, strong vortex ingestion caused pronounced broadband noise modulation known as 'Haystacking', especially at lower inflow veloc…	—	—	—	—	P	—	—	—	—	—	VI	C	simulation
ISO 26101-1:2021 (iTeh preview: Foreword–5.1.5.1.2 only)	"The qualification measurements of the anechoic or hemi-anechoic space shall be made using a bandwidth that is typical of the spectral characteri…	P	—	S	—	—	—	P	—	—	—	VII	C	simulation

SOURCE	KEY FINDING (FROM THE EXTRACTS)	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	LVL	Q	SET
ISO 5305:2024 (iTeh preview: Foreword–7.3.1 only)	"For sound at a high frequency, the significant interference pattern can affect the validity of the far-field condition proposed in Formula (1). …	P	–	P	P	P	–	P	–	P	–	VII	C	simulation
Kanda & Wyss 1986, NBS Technical Note 1305	1 m offset: ±0.8 dB at 250 MHz (OEG), ±2.5 dB at 600–900 MHz, ±3.0 dB at 1000 MHz; at 800 MHz: 0.25 m ±0.6, 0.50 m ±1.2–1.4, 0.75 m ±2.0, 1.00 m ±2.5 …	–	–	P	–	–	–	P	–	–	–	III	C	simulation
Mateljan, ARTA Application Note No. 4	"It is absolutely normal that ripples in a reverberant field are larger than 5dB." (§1, p. 3). "It is recommended that the gate length …	–	–	–	P	–	–	–	P	–	–	VII	C	simulation
Nash 2019, ICA Aachen	Pink-noise fits are near-ideal at 2 kHz (slope −19.8, R^2 0.9) and only "too shallow" at 4–8 kHz (−16.2 … −18.4); the tone data at the same t…	S	–	S	P	–	–	S	–	–	–	III	B	simulation
Okuzono 2024, Appl. Sci. 14:194	Relative error in attenuation ≤0.84 % at 20 kHz (0.05 % with doubled resolution); in the office D < 0.5 dB up to 5 kHz for 20 °C/50 % and 30 °C/70 …	–	–	–	–	–	P	–	–	–	–	VI	A	simulation
Payne & Simmons 1996, NPL Report CIRA(EXT) 009	Measured Lw ranged 90.8–98.4 dB, "a potential range of K2A-factors of 7.6 dB" (§5.1, p. 15). "the variation of sound power level with m…	–	–	S	P	–	–	P	–	–	–	III	B	simulation
Prinn 2023, Acoustics (MDPI) 5:367–395	"An assumption often made in room acoustics simulations is that absorbing surfaces can be described using an LR model. This assumption is not val…	–	–	–	–	–	S	–	–	–	–	VII	A	simulation
Prislan & Svenšek 2017, arXiv 1705.03825	"for the majority of the studied cases the peaks and deeps of the spatially dependent responses coincide remarkably well in frequency, level and …	–	–	–	–	–	S	–	–	–	–	VI	B	simulation
Ressl & Wundes (WSEAS "Recent Advances in Acoustics & Music", ISBN 978	"This raises the cut-off frequency of absorption to 571.6 Hz, and is above the lowest frequency of interest for our research. Even though a certa…	–	P	–	–	–	P	–	P	–	–	VI	C	simulation
Rodriguez (NSI-MI; AMTA-style proceedings, year unclear in text, cites	700 MHz: predicted −34.21 dB vs CST −36.4 dB. Large chamber: measured −16.79/−30.88/−57.63/−52.62 dB vs predicted −11/−27/−55/−55 dB at 100 MHz/400 MH…	–	–	S	–	–	P	–	–	–	–	III	C	simulation
Russo 2018, Euronoise Crete	With the source central, "the lowest one-third-octave band we could measure was 250 Hz band in order to end the traverse 50 cm from the tips of a…	–	–	S	–	–	–	P	P	–	–	III	B	simulation
Schmal, Herrin & Fernández Comesaña (venue/year unclear in text)	Single propeller: dipole BPF (120 Hz) toroid, 2484 and 3328 Hz peaks look monopole-like; "turbulence caused by flow or interactions from the arm …	–	–	–	P	P	–	P	–	–	–	VI	B	simulation
Schmal, Herrin & Fernández Comesaña (year/venue unclear; cites Dec-202	Flow noise "only affects the acoustic results along the negative 90-degree plane … becomes negligible at approximately 450 Hz … masking tonal…	–	–	–	P	P	–	P	–	–	–	VI	B	simulation
Alkmim et al. 2022, JASA 152	Reflection "minimized … but not fully suppressed, especially at low frequencies"; aliasing above 164 Hz	–	–	–	S	–	–	P	P	–	–	III	A	problems
Bahr 2021, NTRS	Closed: reflections/background; open: "severe blurring"; 8 dB residual at 40 kHz	–	–	–	S	–	–	–	–	S	–	VI	B	problems
Bahr, Hutcheson & Stead 2020, NTRS	γ^2 0.2 at 10 kHz open jet vs 0.7 Kevlar; Kevlar noise masks 40 kHz	–	–	–	–	–	–	–	–	S	–	III	B	problems
Belyaev et al. 2015, Acoust. Phys. 61 (RU + EN)	A_avg 2.9–4.6 m^2; α "has physical meaning only for a flat sample"; final judgement by ISO 3745 1/R deviation	–	S	–	–	–	P	–	–	–	–	III	A	problems
Bonfiglio & Pompoli 2013, JASA 134	Agreement at 40–100 Hz; free field survives below cut-off "up to a distance of around 1 m from the boundaries"; λ/4 rule fails above 10 kPa·…	–	S	–	–	–	S	–	P	–	–	III	A	problems

SOURCE	KEY FINDING (FROM THE EXTRACTS)	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	LVL	Q	SET
Brinkmann et al. 2019, JASA 145	Errors "once the assumptions of geometrical acoustics are no longer met"; RT +58 % at 125 Hz; comb filter of a single reflection not matched…	–	–	–	–	–	S	–	–	–	–	III	A	problems
Cunefare et al. 2003, JASA 113	Zero-crossing spacing 29–41 % λ; 0.3 m sampling misses violations at 4 kHz; noise "a marginal indicator"; optimal fit 56→13 % out of toleran…	S	–	S	–	–	–	P	–	–	–	III	A	problems
Fasulo et al. 2025, Aerospace 12	Floor fills nulls at 0/180°; decay ripples beyond 3 D; pylon shifts directivity "up to 4 dB"; no recirculation at 6 D clearance	–	–	–	S	C	–	S	–	–	–	III	A	problems
Friot & Gintz 2009, arXiv	~10 dB at 280 Hz; time windowing "failed" at low frequency; ~100 channels for 100 Hz in 10×7×6 m	–	S	–	–	–	–	–	S	–	–	VI	B	problems
Gallo et al. 2025, Acta Acust. 9	No recirculation rise in 30 s; motor/cooling background; wake vented	–	–	–	–	C	–	–	S	–	–	VI	B	problems
Garg et al. 2019, MAPAN 34	125 Hz: +4.52 → −7.13 dB over 0.64–1.14 m; tolerance only "between 79 and 89 cm"; U = 0.36–0.52 dB	–	S	–	–	–	–	–	P	–	–	III	A	problems
Haasjes 2025, PhD Twente	9.4–9.6 dB reduction to ~600 Hz real-time; chambers "suffer from low-frequency sound waves (up to about 200 Hz) … even above the cut off frequenc…	–	S	–	–	–	S	S	S	–	–	III	A	problems
Hanson et al. 2023, JSV 559 (abstract only)	Reflection zone and shielded zone; "pure acoustic reflection from the ground surface"	–	–	–	S	–	–	–	–	–	–	(abstract)	D	problems
Hochbaum et al. 2026, conf.	55 % of blocks rejected for wind under the drone; BPF band below cut-off for the 12 kg hexacopter	–	S	–	–	–	–	P	–	–	–	III	B	problems
Jawahar et al. 2025, Sci. Rep. 15	Bare plate +4–8 dB above 1 kHz at 50°, absent at 90°; 75PPI-T −13 dB sideline	–	–	–	S	P	–	–	S	–	–	III	A	problems
Jenny & Anderson 2011, JASA-EL 130 (+ BYU thesis)	ORM r_0 13.7/20.6 cm "clearly not accurate"; directivity 22.5 dB off omni; 9.2 λ between points	P	–	S	–	–	–	–	–	–	–	III	B	problems
Jiang, Zhang & Huang 2016, JSV 381	"99% energy absorption means 10% pressure reflection"; effective cut-off 128/120 Hz for 100 Hz design; 57 % of line in tolerance at 100 Hz	–	S	–	–	–	S	–	S	–	–	VI	A	problems
Kim et al. 2022, ICSV28	Upper mics +3 dB from downwash; L_WA 82.6 (lab) vs 82.5 dB (outdoor)	–	–	–	–	–	–	P	–	–	–	III	B	problems
Ma et al. 2022, Appl. Acoust. 185	Absorption ≈ 1 up to ~400 Hz; limit = first excitable cavity mode; "larger … narrower the controllable band"	–	–	–	–	–	–	–	S	–	–	III	A	problems
Ma et al. 2022, Quiet Drones	Adjacent mics differ up to 10 dB at BPF in hemi; 3 dB off 1/R; "could considerably degrade the tonal noise assessment accuracy"	S	–	–	S	–	–	–	–	–	–	III	B	problems
Ma et al. 2024, Acoust. Aust. 52	Onset ~30 s; OASPL unchanged, broadband +1–2 dB, tones broaden; "cannot distinguish the transition" by OASPL	–	–	–	–	P	–	–	–	–	–	III	A	problems
Merino-Martínez et al. 2020, Appl. Acoust. 170	125–200 Hz bands OK only to 1.5 m; jet-column standing wave 125–169 Hz independent of speed; cover nozzle for pure acoustics	–	S	–	–	P	–	S	–	S	–	III	A	problems
Nardari et al. 2019, AIAA	+5 dBA OASPL, +5–10 dB tones ≥ 3rd; standing waves "consequence of the nearly cuboidal shape"	–	S	–	S	S	–	–	–	–	–	VI	B	problems
Palchikovskiy et al. 2016, AIP Conf. Proc.	Free field "within radius of 2 m for tonal noise and 3 m for broadband"; mics ≥ 1 m from wedge tips	S	–	–	–	–	–	S	–	–	–	III	B	problems
Rasmussen 2022, Quiet Drones	First dip c/(4h) ≈ 71 Hz at 1.2 m; dips > 20 dB; "if the source signal contains pure tone components, it will not be possible to do this corre…	S	–	–	S	–	–	–	S	–	–	VI	B	problems
Rizzi et al. 2013, RASD (NASA SALT)	Fails only "close to the room corner … below 250 Hz"; keep mics off walls below 250 Hz	C	–	–	–	–	–	S	–	–	–	III	B	problems
Schlinker & Amiet 1978, NASA CR-145359	Refraction "significant at Mach numbers of 0.1 and greater"; scattering > M 0.2 and > 5 kHz	–	–	–	–	–	–	–	–	S	–	III	B	problems

SOURCE	KEY FINDING (FROM THE EXTRACTS)	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	LVL	Q	SET
Schneider 2009, JSV 320	Local admittance "failed … below ~150 Hz, regardless of the admittance applied"; α 0.997/1.0/0.997 give different regions; asymmetry 170–190…	–	S	–	–	–	S	S	–	–	–	III	A	problems
Simmons, Jobling & Payne 2004, NPL DQL-AC 007	Tone deviations to 14.3 dB while broadband ≤ 3.7 dB; free-r_0 fit passed a bare room; K_2 correction 6.4→0.8 dB; "over-tolerant of poor acoustic ro…	S	–	S	–	–	–	–	–	–	–	III	A	problems
Singh, Garg & Narayanan 2020, IJA 19	312–320 vs 315 Hz "within 3%" on one chamber; slopes 0.01–0.3 dB/doubling at 100 Hz, 3.0–3.7 at 200 Hz	–	S	P	–	–	–	–	–	–	–	VI	B	problems
Stephenson, Weitsman & Zawodny 2019, JASA 145(3) (Letter)	Harmonics ≥ 2 up 15–30 dB after ~7.5 s, BPF unchanged, out of plane; "an artifact of testing within the confined space"	–	–	–	–	S	–	–	–	–	–	III	A	problems
Stroud 2010, AES 129 (abstract only)	Gating limited by driver decay ≥ 30 ms	–	–	–	–	–	–	–	P	–	–	(abstract)	D	problems
Vorländer 2013, JASA 133	JND-accurate RT "requires input data in a quality which is not available from reverberation room measurements"; calibrating by tweaking inpu…	–	–	–	–	–	S	–	–	–	–	VII	A	problems
Wang & Tang 1996, EABE 18	Length and flow resistance dominate; base/air gap barely; 10–20 cm tip cut harmless, 30 cm not	–	P	–	–	–	–	–	S	–	–	VI	B	problems
Wang et al. 2019 (UCMAR), Acta Acust. 105	"distortions rather than a simple source shift"; temperature ratio matters; 12 s vs 1 h	–	–	–	–	–	–	–	–	S	–	VI	A	problems
Weitsman et al. 2020, JASA 148	Clean window 3.9→~10 s with two downstream meshes; upstream mesh worsens; ±1.1→±0.5 dB uncertainty	–	–	–	–	S	–	S	S	–	–	III	A	problems
Whelchel 2023, PhD VT	Harmonics ≥ 7–8 dB with averaging time; screens "no beneficial effect"; ground-board humps 6–8 dB from edge scattering	–	P	–	S	S	–	S	C	P	–	III	A	problems
Winker & Stahnke 2016, Inter-Noise	Free r_0 lets a chamber "fully comply with ISO 3745 … while not exhibiting free-field performance"; λ/2 rule limits small chamber to 214 Hz b…	S	–	S	–	–	–	S	–	–	–	III	B	problems
Zawodny & Haskin 2017, AIAA (LSAWT)	> 5 dB error at one angle at 160 Hz "reflections from the wall-mounted wedges"; λ/4 of tip distance → 175 Hz	–	S	–	–	S	–	S	–	S	–	III	B	problems
acoustic-measurement.com 2025 (vendor)	Cut-off "defined by the acoustic performance of the absorber alone"; lower limit "by the overall … performance of the chamber"	–	S	–	–	–	–	–	–	–	–	VII	C	problems
Hadadi 2024, arXiv:2409.15484 (preprint, BGU / Meta Reality Labs)	PD falls as the number of reflections within 20 ms rises; semi-circular PD significantly below and PFA above em32 (values only in Fig. 2; unclear in t…	–	–	–	–	–	–	–	–	–	S	VI	B	reflection
ISO 26101-2:2024 (ISO/TC 43/SC 1; iTeh preview sample)	K2 corrects energy-averaged surface levels for reflected or absorbed sound, is frequency dependent (K2f per band, K2A overall) and "usually K2 in…	–	–	–	P	–	–	P	–	–	–	VII	C	reflection
Lovedee-Turner 2019, J. Acoust. Soc. Am. 146(5):3339–3352 (accepted ma	Simulated RMS boundary position / dihedral angle (Table I): cuboid 4.63 cm / 8.59°, octagonal 2.69 cm / 2.01°, L 4.69 cm / 14.02°, T 16.45 cm / 8.03°….	–	–	–	P	–	–	P	–	–	S	III	A	reflection
Mabande 2013, J. Acoust. Soc. Am. 134(4):2773–2789	Simulation, J = 4, SNR 30 dB, α = 0.7: mean orientation deviation 1.6°, mean distance deviation 1.52 cm, relative volume error 0.35% (Table II). J = 1…	–	–	–	P	–	–	–	–	–	S	III	A	reflection
Meyer-Kahlen 2022, ICA 2022 Proceedings (Gyeongju; Acoustical Society	Direct sound and floor/ceiling reflections are "easily visible" in SDM energy plots (Fig. 2 caption). Two events inside one analysis block c…	–	–	–	P	–	–	–	–	–	S	VI	B	reflection
Sprunck 2022, arXiv:2208.14017v2 (preprint, Inria / Strasbourg / Cerem	Table I (noiseless): 0–150 image sources recall 94.3%, precision 81.8%, radial error 0.069 mm, angular 0.38°, Euclidean 94 mm, amplitude 0.042; 150–30…	–	–	–	–	–	P	–	–	–	S	VI	B	reflection

SOURCE	KEY FINDING (FROM THE EXTRACTS)	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	LVL	Q	SET
Sun 2012, J. Acoust. Soc. Am. 131(4):2828–2840	EB-MVDR with frequency smoothing + WNG control resolves direct sound plus five first-order reflections and one second-order (walls 2+3) in room 1. Ang...	–	–	–	P	–	–	P	–	–	S	III	A	reflection
Tervo 2013, J. Audio Eng. Soc. 61(1/2):17–28 (Jan/Feb 2013; file label	No localisation error in degrees or cm is reported. Fig. 3 (simulated 20×30×12 m room): "The image-sources with the highest energy are correctly ...	–	–	–	–	–	P	P	–	–	S	VI	B	reflection

02 Convergence summary

THEME	STATEMENT	S (OF WHICH LEVEL III)	P	C	NET	STRENGTH	CONFIDENCE
T1	Pure tones expose deviations that band-noise qualification hides	7 (6)	4	1	+6	Strong	High
T2	Absorber cut-off ≠ chamber usable low-frequency limit	14 (9)	3	0	+14	Strong	High
T3	Qualification method (sampling, fit, standard) changes the verdict	9 (8)	4	0	+9	Strong	High
T4	Ground/floor reflection distorts tonal levels and directivity	10 (6)	13	0	+10	Strong	High
T5	Recirculation in closed rooms inflates rotor harmonics/broadband	5 (4)	7	2	+3	Contested	Medium
T6	α-only / locally-reacting models insufficient; phase-preserving models needed	9 (5)	8	0	+9	Strong	High
T7	Discontinuities and fixtures are the dominant practical reflectors	13 (13)	16	0	+13	Strong	High
T8	Extending free field below passive cut-off is feasible	9 (4)	9	1	+8	Strong	High
T9	Open-jet shear layers refract/scatter; corrections Mach/frequency-limited	6 (4)	2	0	+6	Strong	High
T10	A dominant reflection can be localised from array data	7 (3)	0	0	+7	Strong	High

Counts over all 69 matrix rows; abstract-only rows (Hanson, Stroud) carry no level and are excluded from the Level III count. Strength rule adapted from the skill template: Strong = at least three supporting Level III sources and at most one contradiction; Moderate = two or more supporters; Contested = two or more contradictions, or one against fewer than four supporters. Contradictions are assessed pair by pair in §03.

What the convergent evidence establishes

T2 is the load-bearing finding and the best supported. Fourteen sources across three evidence levels, nine of them Level III, put the chamber's usable limit above the absorber's. The mechanism is consistent: Jiang shows 99 % energy absorption leaves 10 % pressure reflection which "accumulates" near boundaries; Schneider and Haasjes show doors and mounting add to it; Garg and Singh show the readout, a decay slope collapsing from 6 dB per doubling to 0–3 dB and a working space shrinking to 10 cm at 125 Hz. Zawodny adds the operational form: c/(4 × distance to the nearest wedge tip) is the practical floor for any microphone, whatever the room is rated.

T1 and T3 are two faces of one finding. Every controlled study that ran both signals on the same room found tones fail where noise passes (Cunefare, NPL, Nash, Palchikovskiy, Ma 2022); every study that ran two fits on the same traverse found the verdict move (Cunefare, Winker, NPL, Jenny, Russo). The reason is the same in both cases: the deviation is an interference pattern with a spatial period of about a third of a wavelength, and anything that averages over it, in frequency or in fitted distance, erases it.

T4 and T7 say the reflector is usually something you brought. Floors (ten supporting sources, thirteen more touching it), stands and pylons (Fasulo, Weitsman), microphone rigs above 10 kHz (Winker), grates (Nash), doors and recesses (Schneider, ITMO), the array's own distance from the wedges (Zawodny). Only Rizzi's corner failures are attributable to the treatment itself, and only below 250 Hz.

What is strong but qualified

T5, recirculation, is real but its magnitude is geometry-bound; it is the one theme scored Contested. Four Level III sources measure 5–30 dB harmonic growth for stand-mounted rotors in rooms of 3–5 m; two sources measure nothing in 30 s with a vented wake or 6 D of floor clearance; one measures only 1–2 dB broadband for a free-flying vehicle in a 250 m³ room. The disagreement is not about whether it happens but about the ratio of rotor size, thrust and room volume, which no paper models.

T6 is unanimous and better supported than expected. Nine sources agree that absorption-coefficient inputs and locally-reacting boundaries cannot predict the low-frequency field; only Schneider, Brinkmann, Götz and the ITMO group test it against measurement; the rest are simulation or review. The practical consequence is agreed: image-source or wave models with complex, angle-dependent impedance, validated by a traverse.

T8 scores Strong by the counting rule but every supporter is partial in scope. Active control reaches 9–15 dB in a 2-D rig and in simulation (Haasjes), 10 dB at one frequency in a real room (Friot); graded flat-wall linings save 5 % depth (Jiang); meshes buy 6 s of clean rotor data in one room and nothing in another. Every demonstration is small, numerical or single-facility, and the one direct replication attempt (screens, Whelchel) failed; read "Strong" here as "many small yeses", not one large one.

T9 and T10 are well established in their own domains. T9 and irrelevant to a static rig, but it is the reason the aeroacoustic community's "anechoic wind tunnel" numbers do not transfer to a closed chamber: refraction is corrected, scattering above Mach 0.2 and 5 kHz is not. T10, from the reflection-localization set, shows a dominant early reflection can be localised to 1–6° with one compact spherical array at 3–17 dB SNR (Sun), or walls placed to a few centimetres (Mabande, Lovedee-Turner), which is the diagnostic to reach for once a traverse fails.

03 Cross-paper tension inventory

Candidate pairs scoped by shared construct or opposite finding direction (the skill's recall-limited heuristic, not complete pairwise detection). Each resolution is a reading of the texts, offered for the scholar to confirm.

PAIR	TENSION	ASSESSMENT	VERDICT
Weitsman 2020 ↔ Whelchel 2023	Downstream meshes extended the clean window from 3.9 to ~10 s at NASA SHAC; at Virginia Tech "the screens have no beneficial effect on recirculation and turbulence ingestion".	Different rooms (3.25 × 2.55 × 3.86 m sealed vs 4.5 × 2.5 × 2.5 m), rotors (0.24 m vs 0.24 and 0.53 m), mesh spacing and solidity. Whelchel's own reading: "an effective screen configuration is rotor and anechoic chamber dependent". Both Level III, A.	Reconcilable: moderator is facility/rotor geometry. Mesh design does not transfer; must be tested per room.
Nardari 2019 ↔ Ma 2024	Confinement adds ~5 dBA OASPL and 5–10 dB per tone (LBM + hemi-anechoic room) vs "negligible effect on the tonal SPLs" and only 1–2 dB broadband for a hovering Phantom.	Isolated rotor rigidly mounted vs free-flying multirotor whose rotor speeds fluctuate; room 8–12× larger by volume (Ma's own comparison). Energy that a fixed rotor concentrates in harmonics is dispersed around them when RPM wanders; Ma shows broadened, split tones and rising spectral entropy instead.	Reconcilable: same mechanism, different observable. Use tone width and entropy, not OASPL, as the detector for vehicles; harmonic level for stand-mounted rotors.
Stephenson 2019 ↔ Gallo 2025 / Fasulo 2025	Onset within 7.5 s in a sealed ≈32 m³ room (3.25 × 2.55 × 3.86 m) vs no rise in 30 s at VKI and CIRA.	VKI vents the wake through a 1 × 1 m opening into a discharge room; CIRA has ~6 D floor clearance, flow parallel to the floor and > 12 D to the nearest wall along the wake. Gallo also notes the 10 s RPM ramp may hide a transient of similar length.	Reconcilable by geometry; onset time is unmodelled (gap G3).
Cunefare 2003 ↔ Winker 2016 / NPL 2004 / Jenny 2011	Optimal-reference fit "merited" and brings more of the traverse within tolerance vs a free acoustic-centre fit that passes a room with no wedges and yields r_0 = 14–21 cm for a tweeter.	Cunefare bounds r_0: offsets beyond twice the source dimension are flagged and coincide with the worst traverses. The later papers describe the unbounded use permitted by ISO 3745:2003/2012 ("should" fall within 200 mm). ISO 26101 resolved it by fixing the slope.	Reconcilable: the method is sound when r_0 is constrained and reported; the standards' wording allowed the abuse.
Singh 2020 ↔ Jiang 2016	Geometric formula predicts the cut-off "within 3%" vs a modelled effective cut-off 20–28 % above the wedge design value.	Different quantities. Singh's target is the lowest 1/3-octave band with a 6 ± 0.5 dB slope over four discrete distances in one 2.6 × 1.7 × 2.2 m room; Jiang's is 75 % of a centre-to-corner line inside ±1.5 dB at a single tone. Singh's formula is fitted and checked on that one chamber.	Not a contradiction; Singh's precision claim does not generalise (gap G1).
Bonfiglio 2013 ↔ Schneider 2009 / Jiang 2016	Locally-reacting complex image sources give "quite satisfactory" agreement vs local admittance "failed … below ~150 Hz, regardless of the admittance applied".	Bonfiglio compares 1/3-octave band decays along a few radials in 650–800 m³ rooms and reports 1.5–5 dB disagreement within ~1 m of the source; Schneider maps pure-tone 1.5 dB regions on a 0.2 m grid; Jiang locates the local-model error "in the critical region, where the 1.5 dB deviation is violated".	Reconcilable: local models suffice for band-averaged decay, not for tonal pass/fail near boundaries — which is the question that matters here.

PAIR	TENSION	ASSESSMENT	VERDICT
Kim 2022 ↔ Whelchel 2023	Lab and outdoor sound power agree to 0.1 dB vs chamber BPF levels 5–10 dB above outdoor.	A-weighted sound power of a hovering drone at 0.5–1 m over 3–5 m hemispheres vs the BPF tone of a stand-mounted rotor at out-of-plane microphones after 20 s of wake re-ingestion. Kim's metric integrates over the quantities Whelchel's isolates.	Reconcilable: metric and test article differ. Sound power is robust to the room; tonal directivity is not.
Rizzi 2013 ↔ Nash 2019	NASA SALT passes ISO 3745 on pink noise except corners below 250 Hz; Nash's chamber passes on noise and fails on tones.	Rizzi's qualification used noise only. Silence, not contradiction: SALT's tonal status is unreported.	Not assessable; flags that a noise-only pass says nothing about tones (T1).
Rasmussen 2022 ↔ Alkmim 2022 / Jawahar 2025	Pole microphone over a plane cannot be corrected for tones vs foam or porous treatment "mitigates" the reflection.	Rasmussen models the bare and soft-ground cases and shows comb dips remain "considerable" even for 30 kPa·s/m² ground; Alkmim reports the reflection "not fully suppressed, especially at low frequencies"; Jawahar's 24 mm "75PPI-T" foam removes ~13 dB in the 1.5–10 kHz band and nothing below it.	Reconcilable: treatment helps at high frequency; at BPF (100–300 Hz) all three agree it does not, so full anechoic or flush mounting is the only fix.

04 Knowledge gaps

#	GAP	TYPE	IMPLICATION	PRIORITY FOR US
G1	No published tonal qualification of a small chamber (< 60 m³) with a rotor stand and microphone arc in place, traversed along the actual measurement directions as ISO 5305 requires. Palchikovskiy (11.8 × 8.2 × 5.3 m, empty) is the closest.	Empirical	Every number in this literature comes from empty rooms or large facilities; the arc rig's own envelope must be measured, not inferred.	**High**
G2	Room-induced error budget for *directivity*. NPL quantified the sound-power error of a degraded room (0.8–7 dB); no source does the same for elevation directivity of a tonal source.	Methodological	Our deliverable is a polar plot; the uncertainty attached to it has no literature basis yet.	**High**
G3	No model of recirculation onset time or magnitude versus rotor diameter, thrust and room volume; Weitsman and Whelchel both call for one; mesh effectiveness is facility-specific.	Theoretical	Settle times and clean windows must be found empirically per room and per prop.	**High**
G4	Hemi-anechoic vs fully anechoic comparison at BPF for the same rotor exists once (Ma 2022, one drone).	Empirical	The 10 dB adjacent-microphone spread is a single observation; its dependence on height and BPF is unknown.	Medium
G5	Below ~100 Hz: "usual rooms do not allow anechoic measurements below 50Hz" (Friot); large-drone BPF bands fall under the cut-off (Hochbaum). No passive or active solution demonstrated in 3-D at room scale.	Empirical / technological	Two-blade props above 3000 rpm stay above 100 Hz; larger or slower rotors would not.	Low–medium
G6	Only two of ten qualification studies report both signal types and both fit methods on the same data (Cunefare, NPL); most facility papers report one method and one signal.	Methodological	Published "anechoic to X Hz" claims are not comparable across facilities.	Medium
G7	All wedge-absorption and chamber-prediction work is Level VI or single-facility Level III; there is no multi-chamber validation of any cut-off predictor (Singh, Blanco, c/4L).	Empirical	Design formulas are starting points only; simulation with a traverse check is the only validated route (Jiang, Schneider).	Medium

05 What the weight of evidence supports for the arc rig

Graded by the convergence table: **strong** = three or more Level III sources agreeing; **moderate** = fewer or Level VI.

06 Method and limits of this synthesis

Mode: `lit-review` of the deep-research skill (bibliography → source verification → synthesis), executed by hand from the skill files because skills load at session start.

- **Qualify with tones at BPF and its first harmonics, along every arc direction, with the stand and arc installed, against a fixed −6 dB/doubling line.** Strong (T1, T3, T7; Cunefare, NPL, Winker, Nash, Palchikovskiy; ISO 5305 requires it). Report the envelope in Hz × m × direction; a noise-only pass is not evidence for this work.
- **Treat c/(4 d_tip) per microphone and the room's corners below 250 Hz as hard limits, independent of the room's rating.** Strong (T2; Zawodny, Rizzi, Garg, Singh).
- **Expect the room's real limit 20–30 % above the wedge cut-off even in a well-built room; more in a small one.** Strong (T2; Jiang, Schneider, Haasjes, Merino-Martínez).
- **If the floor is hard, either flush-mount or accept that BPF directivity is not measurable; foam under the source fixes only the high band.** Strong (T4; Rasmussen, Ma 2022, Alkmim, Jawahar, Fasulo).
- **Record from spin-up and locate the clean window per run; reverse rotation once per configuration to separate stand asymmetry from prop directivity.** Strong for the signature (T5; Stephenson, Weitsman, Whelchel), moderate for the reversal test (Fasulo alone).
- **Do not use a mesh, screen or settle-time number from another facility.** Strong by contradiction (Weitsman vs Whelchel; G3).
- **If a room is modelled, use complex-impedance image sources or a wave solver with a traverse check; do not use ray tracing or α-only boundaries.** Strong on the negative (T6), moderate on which positive method (Level VI).

Source verification for the eight core papers is logged in `papers/anechoic-simulation/VERIFICATION-2026-09-07.md`; retrieval provenance for the rest in `papers/chamber-problems/RETRIEVAL-A/B.md`. Theme coding and level/quality grades were assigned from the structured extracts, which quote page-referenced passages of the full texts; two sources (Hanson 2023, Stroud 2010) are abstract-only and carry no weight in the convergence counts.

The evidence hierarchy in the skill is medical (RCTs, cohorts). It was mapped to three usable levels for engineering-acoustics literature; all counts should be read with that coarseness in mind. The tension inventory is scoped by shared construct and opposite direction, not exhaustive pairwise comparison. Grey literature (a vendor page, ISO previews) is included as Level VII and never decides a theme on its own.

Corpus bias: the set was assembled around chamber *problems*, so studies reporting unproblematic chambers are under-represented; T2 and T7 counts are therefore upper bounds on how often a chamber disappoints, not estimates of how often it does.

Companion documents: `docs/anechoic-simulation.html` (simulation review), `docs/anechoic-chamber-problems-brief.pdf` (field guide to symptoms), `docs/reflection-localization.html` (locating a failed reflection).
